Econ490: Financial Data Science

Data Science for Macro and Financial Analysis
Department of Economics, Duke University

Instructor Information

Prof. Aguilar || mike.aguilar@duke.edu || https://www.linkedin.com/in/mike-aguilar-econ/
Office Hours: see Canvas; or by appointment

Class Schedule and Delivery

Synchronous, In-Person || Social Science 107
24Aug2026-4Dec2026 || T/R 11:45AM-1:00PM

Pre-Requisites

ECON104D - Econometrics and Data Science
Some familiarity with R and/or Python
Some background in financial economic concepts

Course Description

Financial Data Science is the process of transforming financial data into useful products through statistical modeling, visualization, and software engineering.

This course provides a comprehensive, hands-on introduction to the full data science pipeline as applied to financial and macroeconomic analysis. Students will learn to 1) scope analytic problems, 2) acquire and clean data, 3) engineer features, 4) conduct exploratory data analysis (EDA), 5) construct and evaluate models, 6) visualize results and 7) deploy insights.

Learning Objectives

- Financial Data: Locate, clean, merge, and interpret the major formats of financial data, including market data (OHLCV, returns, bid/ask spreads), fundamental data (financial statements, earnings estimates), macroeconomic data (inflation, yields, employment), and alternative data (news, sentiment, etc..)
- Computing: Use modern data science tools to build reproducible financial applications leveraging tools like R and Python, Git, and APIs
- Visualization and Communication: Design and create appropriate static and interactive visualizations, dashboards, and data storytelling
- Exploratory Data Analysis (EDA): Design and implement appropriate methods to support choice of necessary statistical learning techniques, including univariate statistical distributions, correlations, time dependence, and feature engineering
- Statistical Learning: Design and implement appropriate methods to achieve project goals, including dimension reduction, clustering, regression, classification, and forecasting
- Uncertainty: Implement and interpret techniques such as bootstrapping and Monte Carlo analysis
- Financial Reasoning: Able to explain and apply financial economic concepts such as risk and return, diversification, factor exposures, yields, and volatility

Communication
Assignments, announcements, grades, readings, and other information will be posted on Canvas. The Canvas site will be the primary method of communication for this course, so please check it frequently.

Course Materials
There is no required textbook for this course. Optional readings will be made available throughout the term. Some initial suggestions include:

- Financial Data Science with Python, by Haojun Chen
- Financial Data Science, by Calafiore, El Ghaoui, Fracastoro, and Tsai

Technology

- Cell phones and similar devices: Please do not use cell phones, or similar devices, during class unless otherwise instructed to do so.
- Laptops and tablets: Please bring your laptop to every class. You are welcome to take notes on a tablet. However, the work will be doing together is most conducive to a laptop. Specifically, we will be using R regularly. Be sure to have R available on your laptop. Class demonstrations and solutions will be completed on a Windows OS, but MAC variations are possible.
- Canvas: Course materials such as data, slides, and assignments, will be distributed through the Canvas course site.

Honor Code
I take academic integrity seriously. All students are expected to adhere to the Duke Community Standard. It is the responsibility of each student to understand and follow Duke policies regarding academic integrity. Examples of academic dishonesty include, but are not limited to, cheating on an exam, paraphrasing or summarizing any source without adequate documentation, handing in someone else's work and claiming it is yours (including services such as Coursehero), collaborating with another student inside or outside this class on graded work that is to be done independently, recycling your work or work of other students from other classes. Generative AI may not be used for any class activities or assignments unless otherwise instructed. If I see evidence of any academic dishonesty, I will confront it according to the procedures described in the Bulletin of Duke University.

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- I will not lie, cheat, or steal in my academic endeavors
- I will conduct myself honorably in all my endeavors
- I will act if the Standard is compromised

Grading

Course Project (90%): The vast majority of your course grade will center around a comprehensive capstone-style project. Pretend you are an entrepreneurial financial data scientist. Your task is to develop a product. Specifically, you should identify a macro-financial problem, and then design, implement, and deploy a solution, which you will communicate and critically evaluate.

To ensure you remain on track, your grade will be reflected through weekly deliverables. Each deliverable is designed to support your product development and is timed to correspond to the desired project life cycle.

Class Participation (10%): Participation is judged in several ways.

- Attendance: Active participation is essential in this course. If you miss class, you are responsible for staying up to date and reviewing notes from classmates. Absences due to illness, emergency, or university-related obligations are excused when you submit an Incapacitation Form or Dean's Excuse. Excessive absences, excused or unexcused, may impact your participation grade.
- In-Class Exercises: Most class sessions will culminate with a hands-on exercise that will apply the skills we are exploring that day. I will guide the class through parts of the exercise, and with time permitting, will allow you to work in small groups. You will be able to complete most of the exercise during the allotted class time, but since we all work at different speeds, you will have a post-class window to submit for credit. These submissions are graded for completion, and not for correctness.
- Engagement: Being engaged with the course material is vital for success. A few examples include: Coming to class prepared by completing any necessary readings and pre-work, asking questions in and out of class after having thought carefully about the issue, answering questions I pose to the class, asking your classmates thoughtful questions during presentations, etc..
- Norms: I expect professional behavior and adherence to the Duke Community Standard at all times. Cells phones and similar devices should not be used during class, unless explicitly told to do so. Other examples include, but are not limited to: Being on time and being respectful of classmates. Violations are a forfeiture of your entire Class Participation grade and may be reported to the Office of Student Conduct and Community Standards.

Tentative Schedule

The term *R&D* stands for Research and Development. We will spend the first half of the semester doing Research and the second half in Development.

First Half:

- Week 1: Course expectations and key finance concepts review
- Week 2: Financial data types and sources
- Week 3: Data acquisition via APIs, Git, and code architecture best practices
- Week 4: EDA and Unsupervised Learning
- Week 5: Unsupervised and Supervised Learning
- Week 6: Supervised Learning and Forecasting
- Week 7: Data interfaces such as dashboards and LLMs

Second Half:

The rest of the semester generally will be a mix of status updates, labs, and mini lectures.

- Status Updates: each student (schedule permitting) will get time to present product status updates and gain feedback from their classmates
- Mini Lectures: discussion of data science, finance concepts, or current events that are relevant to products being developed by the students
- Lab: Students will work on product development under the guidance of the instructor

Course Project: Product Development Guidelines

This course is centered around a comprehensive capstone-style project. Pretend you are an entrepreneurial financial data scientist. Your task is to develop a product. Specifically, you should identify a macro-financial problem, and then design, implement, and deploy a solution, which you will communicate and critically evaluate.

1 Product Requirements

Each product must include

- A dashboard built in Streamlit or Shiny and deployed via a cloud service
- At least one interactive visualization
- At least one AI-generated narrative summary
- At least one quantitative analytics engine (e.g., clustering, dimension reduction, regression, classification, simulation, or time series)

Products that appropriately incorporate and deploy more of these elements are generally awarded more points.

You have autonomy to identify the customer base you want to serve and the accompanying problem you want to solve, but it must be in one of the following domains. Other areas will be considered upon request:

- Asset pricing, valuation, or screening
- Portfolio analytics
- Forecasting
- Corporate Fundamentals
- Macroeconomic activity

2 Deliverables

You are responsible for four deliverables related to your product

1. Research Notebook: worth (30%) of assignment grade
2. Development Notebook: worth (55%) of assignment grade
3. Product Repo: worth (10%) of assignment grade
4. Deployed Application: worth (5%) of assignment grade

2.1 Research Notebook

The Research Notebook is intended to document your thinking before implementation begins. It emphasizes problem formulation, product planning, financial and analytical reasoning, and software design. It will be evaluated primarily on the quality of your reasoning, planning, and communication rather than on the eventual success of the final product. This document should be rendered as a single PDF that grows as each section is added throughout the semester.

1. Proposal

   Prepare a brief pitch deck describing the proposed product. At a minimum, identify the target customer, the financial problem being addressed, existing approaches to the problem, your proposed solution, the expected advantages of your solution, and anticipated data sources.

   Products must be approved by instructor before proceeding. Approval will depend upon (i) feasibility within the semester, (ii) alignment with the course learning objectives, and (iii) the potential usefulness of the proposed product.

   Grading will emphasize:

   - Clearly identified target customer and financial problem.
   - A well-motivated and feasible product idea.
   - Appropriate data sources.
   - Professional organization and communication.

2. Requirements Specification

   Provide a written description of what the product should do. Clearly identify the intended users, core features, optional features, expected inputs, expected outputs, and measurable criteria for project success. When appropriate, organize requirements into categories such as *Must Have*, *Should Have*, and *Could Have*. These requirements will later serve as the benchmark for testing and product evaluation.

   Grading will emphasize:

   - Requirements that are specific, measurable, and realistic.
   - Appropriate prioritization of features.
   - Clearly defined inputs, outputs, and success criteria.
   - Consistency with the proposed product.

3. Wireframes

   Prepare preliminary sketches or mock-ups of the proposed user interface. These may be hand-drawn or created digitally. The objective is to communicate the organization of information, user workflow, and major interface components rather than polished visual design. Annotate each wireframe to explain how users will interact with the product.

   Grading will emphasize:

   - Logical user workflow.
   - Clear organization of information and interface components.
   - Consistency with the Requirements Specification.

- Professional presentation and annotation.

4. Finance Domain Guide

   Prepare a two-part overview of the financial and macroeconomic concepts underlying the proposed product. The first part should be a 1–2 page non-technical primer written for users without formal finance training.

   The second part should provide a more technical discussion of the theories, models, market conventions, institutional details, and economic intuition that motivate the product's design and analytical methods. In addition to explaining the relevant financial concepts, discuss the appropriate interpretation of the product's analytical outputs, common misconceptions, important assumptions, limitations, and situations in which the resulting analysis should not be used or overinterpreted.

   This document should be written as a reusable domain guide rather than a traditional report. It should demonstrate your understanding of the underlying finance while also serving as technical documentation for your product. Portions of this guide may later be incorporated into user documentation or used to support AI-generated explanations of your product's results.

   Grading will emphasize:

   - Accuracy, completeness, and depth of the financial discussion.
   - Appropriate communication for both technical and non-technical audiences.
   - Clear connection between the financial concepts and the proposed product, including the interpretation of its analytical outputs.
   - Appropriate discussion of assumptions, limitations, common misconceptions, and situations in which the product's outputs should not be overinterpreted or applied.
   - Organization and clarity of the guide as reusable product documentation.
   - Appropriate use of supporting references.

5. Analytical Methods Guide

   Prepare a two-part overview of the statistical, mathematical, and machine learning methods underlying the proposed product. The first part should provide a 1–2 page non-technical overview written for readers without a quantitative background, explaining the purpose of each analytical method, the types of questions it can answer, and how it contributes to the overall product.

   The second part should provide a more technical discussion of the methodology, mathematical foundations, assumptions, implementation, expected strengths and weaknesses, and the rationale for selecting these methods over reasonable alternatives. In addition to explaining the analytical methods, discuss the appropriate interpretation of the product's analytical outputs, sources of uncertainty, common misconceptions, important assumptions, limitations, and situations in which the resulting analysis should not be used or overinterpreted.

   This document should be written as a reusable analytical guide rather than a traditional report. It should demonstrate your understanding of the underlying analytical methods while also serving as technical documentation for your product. Portions of this guide may later be incorporated into user documentation or used to support AI-generated explanations of your product's results.

   Grading will emphasize:

   - Appropriate selection, justification, and implementation of the analytical methods.

- Accuracy, completeness, and depth of the statistical, mathematical, and machine learning discussion.
- Appropriate communication for both technical and non-technical audiences.
- Clear connection between the analytical methods and the proposed product, including the appropriate interpretation and communication of its analytical outputs and associated uncertainty.
- Appropriate discussion of assumptions, uncertainty, limitations, common misconceptions, and situations in which the product's outputs should not be overinterpreted or applied.
- Organization and clarity of the guide as reusable product documentation.
- Appropriate use of supporting references.

6. System Architecture

   Develop a preliminary blueprint describing how the product will be organized. Include the proposed directory structure, major software modules, expected data flow, external APIs, analytical components, interface components, and interactions among these elements. The emphasis should be on overall software design rather than implementation details. This architecture is expected to evolve as the project develops.

   Grading will emphasize:

   - Logical and modular system organization.
   - Clearly documented data flow and software components.
   - Appropriate decomposition of functionality into reusable modules.
   - Consistency with the Requirements Specification and proposed product.
   - Professional organization and clarity of presentation.

The Research Notebook is intended to document your design decisions before implementation begins. As the semester progresses, previously submitted sections may be revised to reflect instructor feedback or changes in the project's direction. Consistency across notebook sections is expected; when major design decisions change, earlier sections should be updated accordingly.

The Research Notebook is evaluated primarily on the quality of your reasoning rather than the eventual success of the project. Students are expected to justify design decisions, demonstrate thoughtful planning, and show a clear understanding of the financial and analytical concepts that motivate the proposed product. The notebook is intended to document your thinking before implementation begins. AI tools may be used to improve grammar and organization, but not as a substitute for the substantive content of the notebook.

2.2 Development Notebook

The Development Notebook documents the implementation and evolution of your product throughout the semester. Unlike the Research Notebook, which emphasizes planning before implementation, the Development Notebook records the engineering process itself. It should function as a living design notebook that explains how the product evolved, documents technical decisions, records supporting evidence, and reflects upon challenges encountered during development.

The notebook should be rendered as a single PDF that grows throughout the semester. Updated versions will be submitted as development milestones are completed. Because product development is inherently iterative, previously completed sections should be revised whenever improved understanding, instructor feedback, or changing design decisions warrant modification.

The notebook should be organized as follows:

1. Project Snapshot
2. Data Acquisition
3. Data Preparation
4. Exploratory Data Analysis
5. Feature Engineering
6. Analytics Engine
7. User Interface
8. Testing and Validation
9. Deployment
10. Product Evaluation
11. AI Usage Log

The Development Notebook should communicate not only what was built, but also how and why the product evolved throughout the semester. Earlier milestones should be revised whenever necessary so that the notebook remains internally consistent and accurately reflects the current state of the project.

2.2.1 Project Snapshot

The notebook should begin with a one-page Project Snapshot that is updated throughout the semester. At a minimum, it should include

- Product name
- Target customer
- Problem statement
- Repository link
- Deployment link
- Current project status

2.2.2 Data Acquisition

Purpose: Acquire and document the data required to build the product.

Tasks

- Identify and justify the selected data sources.
- Obtain the data using APIs, downloads, databases, web scraping, or other appropriate methods.
- Document variables, sample period, data frequency, and licensing restrictions.
- Evaluate data quality, completeness, and important limitations.

Documentation: Include representative code, tables, screenshots, workflow diagrams, or other evidence demonstrating successful acquisition.

Grading will emphasize

- Appropriate selection and justification of data sources.
- Complete documentation of the acquisition process.
- Understanding of data quality and limitations.
- Appropriate supporting evidence.
- Professional organization and communication.

2.2.3 Data Preparation

Purpose: Transform the raw data into analysis-ready data.

Tasks

- Clean, merge, reshape, and transform the data.
- Handle missing values and other quality issues.
- Apply normalization, scaling, or other preprocessing where appropriate.
- Justify important preprocessing decisions.

Documentation: Include representative code, before-and-after summaries, tables, or figures documenting the preparation process.

Grading will emphasize

- Thorough and reproducible data preparation.
- Appropriate preprocessing decisions.

- Recognition of downstream implications.
- Appropriate supporting evidence.
- Professional organization and communication.

2.2.4 Exploratory Data Analysis

Purpose: Develop an understanding of the data and identify findings that influence later development.

Tasks

- Produce appropriate descriptive statistics.
- Explore the data using visualizations.
- Identify important patterns, relationships, and anomalies.
- Explain how the exploratory analysis informed subsequent decisions.

Documentation: Include representative figures, tables, statistical summaries, and explanatory discussion.

Grading will emphasize

- Insightful exploration.
- Appropriate visualizations and summaries.
- Connection between findings and later development.
- Appropriate supporting evidence.
- Professional organization and communication.

2.2.5 Feature Engineering

Purpose: Construct or select variables that improve the effectiveness of the analytical methods.

Tasks

- Create engineered features.
- Perform feature selection or dimensionality reduction where appropriate.
- Justify major engineering decisions.
- Discuss alternatives considered.

Documentation: Include representative code, comparison tables, diagnostic plots, or other supporting material.

Grading will emphasize

- Thoughtful feature engineering.
- Appropriate justification of design decisions.
- Technical correctness.
- Appropriate supporting evidence.
- Professional organization and communication.

2.2.6 Analytics Engine

Purpose: Develop the statistical, machine learning, optimization, forecasting, simulation, or other analytical methods that generate the product's core outputs.

Tasks

- Select and justify the analytical methodology.
- Train, tune, and evaluate models where appropriate.
- Interpret analytical outputs.
- Compare with reasonable alternatives when applicable.

Documentation: Include representative code, evaluation metrics, diagnostics, validation results, and discussion.

Grading will emphasize

- Appropriate methodology.
- Technical correctness.
- Model evaluation and interpretation.
- Appropriate supporting evidence.
- Professional organization and communication.

2.2.7 User Interface

Purpose: Design and implement the interface through which users interact with the product.

Tasks

- Develop the user workflow.
- Implement visualizations and interactive controls.
- Communicate analytical results clearly.

- Refine usability.

Documentation: Include screenshots, interface diagrams, or walkthroughs illustrating the user experience.

Grading will emphasize

- Usability and interface design.
- Effective communication of analytical results.
- Appropriate supporting evidence.
- Professional organization and communication.

2.2.8 Testing and Validation

Purpose: Demonstrate that the product functions correctly and produces reliable results.

Tasks

- Test functionality.
- Validate analytical methods.
- Evaluate robustness and edge cases.
- Document bug fixes and remaining limitations.

Documentation: Include test results, validation metrics, bug logs, or other supporting evidence.

Grading will emphasize

- Thorough testing.
- Appropriate validation.
- Recognition of remaining limitations.
- Appropriate supporting evidence.
- Professional organization and communication.

2.2.9 Deployment

Purpose: Deploy the completed product so it is accessible to users.

Tasks

- Configure the deployment environment.
- Manage software dependencies.

- Document installation and deployment procedures.
- Address deployment challenges.

Documentation: Include deployment screenshots, logs, links, or workflow summaries.

Grading will emphasize

- Successful deployment.
- Reproducibility.
- Appropriate supporting evidence.
- Professional organization and communication.

2.2.10 Product Evaluation

Purpose: Evaluate the completed product relative to the original customer problem and project requirements.

Tasks

- Assess strengths and weaknesses.
- Discuss assumptions and limitations.
- Identify appropriate use cases.
- Propose future improvements and reflect on lessons learned.

Documentation: Support your evaluation using comparisons to project requirements, user feedback, performance summaries, or other appropriate evidence.

Grading will emphasize

- Thoughtful evaluation.
- Honest discussion of limitations.
- Appropriate supporting evidence.
- Professional organization and communication.

2.2.11 AI Usage

You are permitted to use AI tools to assist with all aspects of coding, from debugging to full code generation. You may use AI tools (e.g. Chat GPT, Claude, etc..) either via web, CLI, or integrated within your IDE. You are encouraged to use AI as a software development assistant—for brainstorming, explaining concepts, debugging, code generation, and improving documentation. However, you remain responsible for all technical decisions, interpretations, conclusions, and submitted work. AI should support your thinking, not replace it.

To support your responsible usage of AI, the final section of your Development Notebook should be an AI-usage log where you document substantive interactions with AI. Your log should have the following structure:

- **Entry 1**
  - **Date:** Sept. 18
  - **Milestone:** Feature Engineering
  - **Task:** Compute rolling beta.
  - **AI Assistance:** ChatGPT suggested an implementation using `slider::slide()`.
  - **Verification and Revisions:** Compared results against manual calculations on several securities, corrected the window alignment, and confirmed the implementation produced the expected results.
  - **Outcome:** Incorporated a modified version of the AI-generated code into the project.
- **Entry 2**
  - **Date:** Oct. 2
  - **Milestone:** Exploratory Data Analysis
  - **Task:** Identify informative visualizations for the distribution of daily returns.
  - **AI Assistance:** ChatGPT suggested several visualization approaches and provided code for a histogram with an overlaid density curve.
  - **Verification and Revisions:** Reviewed the code, confirmed that the plotted values matched the underlying data, and modified the plot aesthetics and axis labels for clarity.
  - **Outcome:** Used the revised visualization in the development notebook and final application.

Grading will emphasize:

- Appropriate use of AI as a tool to support, rather than replace, the student's own reasoning, decision making, and understanding.
- Clear documentation of substantive AI interactions that materially influenced the project's design, implementation, analysis, or written content.
- Evidence that AI-generated outputs were critically evaluated, tested, verified, and, when appropriate, revised before incorporation into the project.
- Professional reflection on how AI contributed to the development process, including instances where AI suggestions were modified, rejected, or corrected.

The AI Usage Log is intended to document your development process, not to discourage AI use. Thoughtful and well-verified use of AI is encouraged; the quality of your engineering judgment, not the quantity of AI usage, will be evaluated.

2.3 Product Repository

- This is your software product and supporting code base.
- The repository must be maintained using Git and submitted via GitHub (details provided in class).
- The repository must contain a README describing the project, directory organization, software dependencies, installation instructions, and directions for reproducing and running the product.
- Code should be organized into reusable, well-documented modules and functions rather than a single monolithic script.
- Data Management: Students are expected to manage data using practices appropriate for professional data science workflows rather than treating large datasets as source code.
  - Small static datasets may be included directly in the GitHub repository.
  - Medium-sized datasets should generally be stored using efficient binary formats (e.g. `parquet`, `fst`, or `RDS`) rather than CSV whenever practical.
  - Large raw datasets should not be committed to GitHub. Instead, the application should retrieve data from an external source, use a cached copy, or download the data during initialization.
  - Applications that rely on public APIs (e.g. FRED, Alpha Vantage, Polygon, Yahoo Finance, etc.) should minimize unnecessary API requests by caching downloaded data locally and refreshing the cache only when appropriate.
  - The repository should include sufficient code and documentation to reproduce all required datasets from their original source whenever feasible.
  - The instructor must be able to run the repository without requiring proprietary datasets or paid data services unless prior approval has been obtained.
- Grading will emphasize:
  - Organization and architecture of the code base
  - Readability, documentation, and maintainability
  - Appropriate use of functions, packages, and modular design
  - Effective use of version control and meaningful commit history
  - Reproducibility of the project from the repository
  - Appropriate management of datasets and external data sources

2.4 Deployed Application

- This is the deployed software product that users interact with.
- The application should be accessible to the instructor and function as described in the Research Notebook.
- Applications should load data efficiently. Long-running downloads or repeated API calls should be avoided through appropriate use of cached data or scheduled data refreshes.

- Grading will emphasize
  - Functionality and reliability
  - User experience and interface design
  - Appropriate communication of financial results
  - Responsiveness and robustness
  - Professional appearance and usability (e.g. clear labels, readable plots, etc..)

3 Submission Instructions

- Due dates and times for all deliverables will be indicated on the course Canvas site
- The deliverables associated with the Research Notebook and Development Notebook will be submitted via Canvas
- The Product Repo will live on each student's Class Github repo (details forthcoming)
- Links to the Deployed Application will be included within the Product Snapshot at the beginning of your Development Notebook.

Other Disclosures:

Modifications: The instructor reserves the right to alter any of the aforementioned components of this syllabus or other class related materials in an effort to enhance the pedagogical experience.

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- I will not lie, cheat, or steal in my academic endeavors
- I will conduct myself honorably in all my endeavors
- I will act if the Standard is compromised

Academic Accommodations: If you are a student with a disability and need accommodations for this class, it is your responsibility to register with the Student Disability Access Office (SDAO) and provide them with documentation of your disability. The SDAO will work with you to determine what accommodations are appropriate for your situation. Please note that accommodations cannot be provided until the instructor receives a Faculty Accommodation Letter from the SDAO. Contact: sdao@duke.edu

Resources to Support Student Success: If you are having trouble completing assignments or understanding the materials, please consult with me about appropriate course preparation and readiness strategies as needed. Either send me an email or visit office hours to discuss the personal or academic difficulties you are facing. I may also direct you to other resources on campus.

The Academic Resource Center (ARC) offers services to support students academically during their undergraduate careers at Duke. The ARC can provide support with time management, academic skills and strategies, course-specific tutoring, ADHD/LD coaching, and more. ARC services are available free to any Duke undergraduate students, studying any discipline. Contact: (919) 684-5917 or theARC@duke.edu

Mental Health and Wellness Resources: Student mental health and wellness are of primary importance at Duke, and the university offers resources to support students in managing daily stress and self-care. Some resources are listed below:

DuWell provides Moments of Mindfulness (stress management and resilience building) and meditation programming (Koru workshop) to assist students in developing a daily emotional well-being practice. All are welcome and no experience is necessary.

If your mental health concerns and/or stressful events negatively affect your daily emotional state, academic performance, or ability to participate in your daily activities, many resources are available to help you through difficult times.

DukeReach provides services for students who are experiencing significant challenges relating to mental health, physical health, social adjustment, and/or a variety of other stressors.

Counseling and Psychological Services (CAPS) services include individual and group counseling services, psychiatric services, and workshops. To initiate services, walk-in/call-in 9:00 AM – 4:00 PM (M/W/Th/F) and 9:00 AM – 6:00 PM Tuesdays. CAPS also provides referral to off-campus resources for specialized care. Contact: (919) 660-1000

TimelyCare (formally known as Blue Devils Care) is an online platform that is a convenient, confidential, and free way for Duke students to receive 24/7 mental health support through TalkNow and scheduled counseling.

Financial Support: If you are having difficulty with textbook, software, or supply costs associated with this course, here are some resources:

- Contact the financial aid office (whether you are on aid or not). They have loans and resources for connecting students with programs on campus that might be able to help alleviate these costs.
- DukeLIFE offers course materials assistance for eligible students. Please note that students who are eligible for DukeLIFE benefits are notified prior to the start of the semester; program resources are limited. Students who have limited access to computers may request loaner laptops through the DukeLIFE Technology Assistance Program. Please note that supplies are limited.
- Duke Libraries offers textbook rentals through the Top Textbooks at the Duke Libraries, where you can rent a textbook for 3 hours at a time.

Discussion Guidelines: Civility is an essential ingredient for academic discourse. Differences in beliefs, opinions, and approaches are to be expected. Active interaction with peers and your instructor is essential to success in this course, so please adhere to these guidelines:

- Respect that others' opinions and beliefs may differ from your own. If you disagree, you may critique the idea, but not the person.
- Listen carefully, be courteous and don't interrupt.
- Support your statements with evidence and a rationale.
- Try to moderate how you contribute to the discussion – if you have a lot to say, try to avoid dominating the conversation; if you are reluctant to speak up, try to find an opportunity to share your perspective.

Please bring any communications you believe to be in violation of this class policy to the attention of your instructor.

